

NOT MEASUREMENT
SENSITIVE

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SUPERSEDING
CRFD/PD-06-26

PURCHASE DESCRIPTION

BOOT, COMBAT, INTERMEDIATE COLD WET

This Purchase Description is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This Purchase Description covers the requirements for a Desert Sand colored leather, flesh side out, combat boot with a direct attached soling system for use in cold weather environments. The boot will have a mildew resistant and waterproof upper, plain toe, partial speed lace closure system, and padded collar. Each pair of boots will have two (2) pair of removable, insulated booties.

1.2 Classification. The boot will be available in the following sizes and widths as specified.

1.2.1 Schedule of sizes and widths.

Size: 3 through 15 (whole and half sizes)
16 (whole size only)

Widths:

Narrow (B)	Wide (EE)
Regular (D)	X-Wide (EEEE)

The removable insulated bootie will be in whole sizes only, and in two (2) widths, one size fitting Narrow and Regular widths and the other fitting Wide and Extra Wide widths. The size of the boots will be determined by the size of the entire boot/insulated bootie combination when worn together for each respective size. Each size of insulated bootie will also fit a boot that is 1/2 size larger.

Comments, suggestions, or questions on this document should be addressed to: DLA Troop Support, Attn: DSCP Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096 Since contact information can change, you may want to verify the currency of this address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database at <https://assist.dla.mil>.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this purchase description. This section does not include documents in other sections of this purchase description or recommended for additional information as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents listed in sections 3, 4, or 5 of this purchase description, whether they are listed or not.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those noted in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-55093	- Laces, Nylon
A-A-55126	- Fastener Tapes, Hook and Loop, Synthetic

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-3122	- Leather, Cattlehide, for Footwear Uppers and Gussets, Chrome Tanned, Fatliquored
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(Copies of these documents are available online at <https://assist.dla.mil/quicksearch/> or from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

DEFENSE LOGISTICS AGENCY TROOP SUPPORT (DLA TS)

2-1-1635	- Speed Lace Assembly
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(Copies of drawings are available from the DLA Troop Support, Philadelphia, 700 Robbins Avenue, 19111-5096, ATTN: DSCP-CRFD-TDR.)

ENVIRONMENTAL PROTECTION AGENCY

Regulations for the Enforcement of the Federal Insecticide, Fungicide and Rodenticide Act
(40 CFR Part 162)

(Copies are available online at <http://www.epa.gov/pesticides> or from the Environmental Protection Agency, 1200 Pennsylvania Avenue, N.W., Washington, DC 20460.)

2.3 Non-Government standards and other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Test Method 20A – Fiber Analysis Quantitative
AATCC Test Method 100 - Assessment of Antibacterial Finishes on Textile Materials
AATCC Test Method 127 - Water Resistance: Hydrostatic Pressure Test
AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles

(Copies of are available on line at <http://www.aatcc.org/> or from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection of Attributes

(Copies are available online at <http://www.asq.org> or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

AMERICAN TYPE CULTURE COLLECTION (ATCC)

ATCC – 6538 - Staphylococcus Aureus
ATCC – 9027 - Pseudomonas Aeruginosa

(Information available online at <http://www.atcc.org> or from the American Type Culture Collection (ATCC), P.O. Box 1549, Manassas VA 20108 USA.)

ASTM INTERNATIONAL

ASTM C518 - Standard Test Method for Steady-State Thermal Transmission Properties
by Means of the Heat Flow Meter Apparatus
ASTM D297 - Standard Test Methods for Rubber Products Chemical Analysis
ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic
Elastomers Tension
ASTM D471 - Standard Test Method for Rubber Property Effect of Liquids
ASTM D624 - Standard Test Method for Tear Strength of Conventional Vulcanized
Rubber and Thermoplastic Elastomers

- ASTM D751 - Standard Test Methods for Coated Fabrics
- ASTM D1052 - Standard Test Method for Measuring Rubber Deterioration Cut Growth Using Ross Flexing Apparatus
- ASTM D1056 - Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber
- ASTM D1630 - Standard Test Method for Rubber Property - Abrasion Resistance (Footwear Abrader)
- ASTM D1777 - Standard Test Method for Thickness of Textile Materials
- ASTM D2208 - Standard Test Method for Breaking Strength of Leather by the Grab Method
- ASTM D2240 - Standard Test Method for Rubber Property - Durometer Hardness
- ASTM D3273 - Standard Test Methods for Resistance to Growth of Mold
- ASTM D3574 - Standard Test Methods for Flexible Cellular Materials - Slab, Bonded, and Molded Urethane Foams
- ASTM D3776 - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
- ASTM D3787 - Standard Test Method for Bursting Strength of Textiles - Constant-Rate-of-Traverse (CRT) Ball Burst Test
- ASTM D3886 - Standard Test Method for Abrasion Resistance of Textile Fabrics (Inflated Diaphragm Apparatus)
- ASTM D4786 - Standard Test Method for Stitch Tear Strength, Single Hole
- ASTM D4966 - Standard Test Method for Abrasion Resistance of Textile Fabrics (Martindale Abrasion Tester Method)
- ASTM E96 - Standard Test Methods for Water Vapor Transmission of Materials
- ASTM F392 - Standard Practice for Conditioning Flexible Barrier Materials for Flex Durability
- ASTM F1614 - Standard Test Method for Shock Attenuating Properties of Materials Systems for Athletic Footwear

(Copies of documents are available online at <http://www.astm.org> or from the ASTM INTERNATIONAL, 100 Barr Harbor Drive, P.O. Box 700C West Conshohocken, PA 19428-2959.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA-1971 Standard on Protective Ensemble for Structural Firefighting (2007 Edition)

(Copies of this document are available online at <http://www.nfpa.org> or National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02169-7471.)

SATRA GLOBAL TEST METHODS

- SATRA TM2 - Tensile properties of insole materials
- SATRA TM3 - Flexing index
- SATRA TM14 - Resistance to scuffing by mild circular abrasion
- SATRA TM31 - Abrasion Resistance – Martindale method
- SATRA TM64 - Compression Set – constant stress method

SATRA TM77/MTM77	- Flexing machine - water penetration test
SATRA TM80	- Transverse tensile strength of sheet materials
SATRA TM83	- Measurement of the area shape retention and collapsing load of formed toe puff and stiffener materials
SATRA TM98	- Dimensional stability with changes in atmospheric humidity
SATRA TM101	- Surface peel strength of insole materials
SATRA TM144	- Friction (slip resistance) of footwear and floorings
SATRA TM223	- Floor marking by solings or top pieces
SATRA STM184H	- Whole Shoe Flexing Machine – High leg
SATRA STD185	- Sole Adhesion Tester

(Application for copies should be addressed to <http://info@satra.co.uk> or from the SATRA Technology Centre, Wyndham Way, Telford Way, Kettering, Northants ; NN 16 8SD, United Kingdom)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Terms.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.1.2 Standard sample. The finished leather shall match a swatch of cattlehide leather for shade and sueded flesh appearance and unless otherwise indicated, shall be equal to or better than the sample with respect to all characteristics for which the standard is referenced.

3.1.3 Visual shade matching (All components). The color and appearance of the components shall match the standard for shade and appearance (see 3.1.2) when tested as specified in 4.4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled recovered or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the requirements of this document and promotes economically advantageous life cycle costs.

3.2.1 Leather.

3.2.1.1 Upper leather. The leather shall be flesh out drum dyed Desert Sand with color penetrating throughout the thickness of the hide so that the chrome tannage is not visible. The leather for the vamps, quarters, backstay/pull-tab shall conform to MIL-PRF-3122, Type I, Treatment A and Treatment B. The leather for the eyelet stays; top band and gusset/tongue shall conform to MIL-PRF-3122, Type II, Treatment A. The following exceptions to MIL-PRF-3122 shall be applicable for the upper leathers:

- a. The flesh surface shall be sueded as to produce a fine uniform nap.
- b. The grain surface shall be lightly buffed to remove the surface of the grain only.
- c. The finish requirements for the leather shall not apply.
- d. The tear strength requirements for the eyelet stays and top band shall not apply. Eighty percent (80%) of the gusset/tongue leather samples must meet a minimum of 23 pounds stitch tear.
- e. The thickness shall be as specified for the cut parts in Table IX.
- f. As an option, the eyelet stays may conform to MIL-PRF-3122, Type I, Treatment A and Treatment B.

3.2.2 Insoles. The insole shall meet the requirements specified in Table I when tested as specified in 4.4.2.

TABLE I. Insole performance requirements

Characteristics	Physical Requirements
Thickness, inches (min.)	0.090
Fungal resistance, Level	10
Wet Tensile, kgf/cm ² (min.)	70
Flexing Index, (5000 cycles) (min.)	3.7
Abrasion Resistance	
Cellulose board, mm ³ (max.)	50
Non-Woven board, revs dry/wet (min.)	25,600 /6400
Transverse Tensile-Dry, kgf/cm ² (min.)	8
Dimensional Stability, % (max.)	0.7
Peel Strength, kgf/cm ² (min.)	0.5

3.2.3 Fabrics.

3.2.3.1 Non removable boot lining. The boot lining shall be constructed from pattern pieces cut from waterproof 3-layer laminate material. The three-layer laminate lining shall consist of a textile face cloth, a waterproof film, and a tricot knit cloth and shall conform to the requirements specified in Table II, when tested as specified in 4.4.2. The face layer shall consist of a textile that meets the Martindale Abrasion, Burst Strength and Bacterial Resistance performance listed in Table II when tested as specified in 4.4.2. Any antimicrobial technology used shall be an EPA-registered antimicrobial and shall be durable for the life of the item. The requirements of this paragraph shall apply to all lining within

the boot including the fabric top cover (face cloth) attached to the removable cushioned insert. The waterproof film shall be polytetrafluorethylene, microporous (PTFE). The tricot knit cloth shall be nylon, weighing a minimum of 1.0 ounce per square yard and meet the requirements in Table II when tested as specified in 4.4.2 as part of the three-layer laminate lining specified. The color of the tricot knit cloth shall be natural unless otherwise specified in the contract or solicitation.

TABLE II. Laminated cloth performance requirements

Characteristics	Requirements
Martindale Abrasion (Fabric only) Dry (153,600 revs), cycles (min)	9,600
Bursting Strength (lbs.) Fabric Only (min)	125
Bacterial Resistance , percent reduction (min) Staphylococcus Aureus Pseudomonas Aeruginosa	99 99
Hydrostatic Resistance, pounds /square inch (psi) (Liner cloth) Initial (min) At seam (min) After synthetic perspiration (min) After diesel fuel oil (minimum)	50 50 50 50
Water Permeability/ Resistance (liner cloth) Initial (min) At seam (min) After synthetic perspiration (min) After diesel fuel oil (min)	No leakage No leakage No leakage No leakage
Abrasive Durability (600 cycles) Back / Leather Side (min)	No leakage
Moisture Vapor Transmission Rate (MVTR), g/m ² /24hrs B method (min) BW method (min)	640 4,000
Flex Durability Gelbo (3,000 cycles) (min) Wet, (24 hours to leak) (min) Wet, (96 hours to delamination) (min)	No leakage No leakage No delamination

3.2.3.2 Color. The color of the face side of the laminated cloth shall be gray based on Government approval unless otherwise specified in the contract or solicitation.

3.2.4 Removable insulated booties. The removable insulated booties shall have an insulated batting of 200 (± 10) grams per square meter of performance insulation as defined in Table IIIA. The insulated layer shall be sandwiched between two (2) outer layers of fabric, which approximate gray herringbone in color (or equal based on Government approval) and meet the Martindale abrasion and burst strength performance criteria listed in Table II when tested as specified in 4.4.2. The quilted lining construction shall conform to the requirements of Table IIIB when tested as specified in 4.4.2.

TABLE IIIA. Performance insulation batting specifications

Characteristics	Minimum	Maximum
Composition (polyester %)	10	100
Weight (grams/sq.m)	190	210
Thermal Resistance R-Value (h.F.ft ² /Btu)	0.27	N/A
Water Gain (wt%)	N/A	95

TABLE IIIB. Insulated bootie quilt requirements

Characteristics	Minimum	Maximum
Weight (grams/sq.m)	332	N/A
Thermal Resistance (clo)	0.4	N/A
Thickness (inch)	0.098	N/A

3.2.5 Box toe. The box toes shall be skived 3/8 ($\pm 1/8$) inch wide at the breast. A Thermoplastic box toe may be used and shall conform to the performance requirements of Table IV when tested as specified in 4.4.2.

TABLE IV. Thermoplastic performance requirements

Characteristics	Physical Requirements
Initial Collapsing Load (N)	>130
Resilience (%) (min)	25
Moisture Resistance (%) (min)	80
Area Shape Retention (%) (min)	
Initial	85
10 th collapse	60

3.2.6 Counters. An extruded thermoplastic film sandwiched between fusion bonded non-woven fabric made from a blend of synthetic fibers and impregnated with a styrene co-polymer

and coated on each side with a thermoplastic adhesive may be used. A Thermoplastic counter may be used and shall conform to the performance requirements of Table IV when tested as specified in 4.4.2.

3.2.7 Shank. The fiberglass shanks shall be bonded to the insole and shall conform to the shape of the last. The fiberglass shank (glass and resin area), when cured, shall be 5/8 ($\pm$ 1/16) inch in width.

3.2.8 Speed lace assembly. The speed lace assembly shall be brass, with a coated nylon finish in accordance with Drawing 2-1-1635. On each quarter, there shall be five (5) speed lace loops for sizes 3 through 6-1/2, and six (6) for sizes 7 through 15. The color shall be Desert Sand.

3.2.9 Eyelets. The two bottom eyelets shall be brass, with a coated nylon gloss finish, size AA. The color shall be Desert Sand.

3.2.10 Removable cushion insert. The removable cushion insert shall be a low-density polyurethane heel cup support made of a pure polyether polyurethane system, medium black in color, and bonded to a cloth, meeting the face cloth performance requirements in Table II. The total thickness of the insert shall be a minimum of 0.100-inch forward of the instep, and a minimum of 0.220-inch elsewhere. The cushion insert shall be mated by size to the correct size boot and shall be uniform. Testing shall be as specified in 4.4.2.

TABLE V. Removable cushion insert requirements

Characteristics	Physical Requirements
Density, lb/cu. ft.	16-19
Foam Hardness, Shore 00	50-65
Tensile strength, psi	>82
Compression set @ 25%, percent (max)	<5
Compression set @ 50%, percent (max)	<5

3.2.11 Laces. The laces shall conform to Type III, Class 1 of A-A-55093. The color shall be Desert Sand. The length of the laces shall be a minimum of 72 inches for sizes 3 through 6-1/2, 87-inches for sizes 7 through 11, and 97-inches for sizes 11-1/2 through 16.

3.2.12 Fastener tape. The fastener tape for the removable bootie shall be black and a minimum dimension of 3/4-inches by 3-inches. The pile fastener shall conform to Class 1 of A-A-55126. The hook fastener tape shall conform to Type I, Class 1 of A-A-55126 (see Figure 3B).

3.2.13 Collar foam. The foam rubber for the padded collar shall be 1/4 ($\pm$ 1/32) inches thick. The foam rubber shall have a compressibility of 2 to 5 pounds per square inch (psi) (defined as compression to deflect 25%) when tested in accordance 4.4.2.

3.3 Soling system requirements.

3.3.1 Midsole construction and design. The cushion midsole shall be directly attached to the upper and shall be achieved either through an injection molding or open pouring method. The design shall be as shown in Figure 3A. The cushion midsole shall be pure polyether polyurethane, Desert Sand color and have an exterior texture pattern that is a good match to the Government standard sample (see 6.4). The dimensions of the cushion midsole shall be sufficient to meet all of the salient characteristics listed within this document.

3.3.2 Midsole compound requirements. The requirements for the pure polyether polyurethane cushioned midsole are based on testing done on 1/2-inch test slabs for all midsole tests except Ross flex which uses 1/4-inch slabs (according to ASTM procedures) and not actual midsoles. The 1/4-inch test slab is combined with a 2.0mm insole board and conditioned at -5°C for 24 hours prior to Ross flex testing. The requirements shall be as specified in Table VI when tested in accordance with 4.4.2.

TABLE VI. Midsole polyether polyurethane requirements

Characteristic	Minimum	Maximum
Density gm/cc	0.40	0.60
Hardness shore A	50	75
Tensile strength kgf/cm sq	25	N/A
Elongation, Percent	250	N/A
Tear Strength Die T kg/cm	3.7	N/A
Volume Swell, Percent IRM 903 (at 24 hrs)	N/A	10
Compression Set (Percent)	N/A	20
Ross Flex (1-inch X 6-inch) at -5 °C (Percent) 150,000 cycles	N/A	900

3.3.3 Outersole. The outersole design shall be the Quabaug Vibram 360-Degree Lug Model (see Figure 4). The outsole shall be a minimum of 18-iron (9.5 millimeters) in thickness of solid rubber at each of the lugs. The color of the rubber outsole including any identification plug shall be Desert Sand in color.

3.3.3.1 Rubber outersole compound requirements. The requirements for the rubber outsole are based on testing done on slabs (according to ASTM procedures) and are not actual soles. The requirements shall be as specified in Table VII when tested in accordance with 4.4.2.

TABLE VII. Rubber outsole compound requirements

Characteristics	Outsole
NBS abrasion (min)	120
Hardness (shore A)	57-67
Stitch Tear, pounds (min)	
Dry	300
Fuel B	50
Die C tear strength, pounds (min)	250
Ross flex, Oil #3, percent (max)	
50,000 flexes	250
Volume swell, Percent (max)	
Fuel B (46 hrs)	70
Non-marking	Pass
Tensile, Pounds (min)	2,100
Elongation, Percent (min)	400
Cold Flex, -23 °C percent (max)	
7,500 flexes	<500
Transitional Hardness (max change)	
After 2 hrs at -23°C	+10 point shore hardness
Slip Index, rough/ice (min)	0.20

3.3.4 Shock attenuation and kinematic parameters. Whole boot shock attenuation as measured in accordance with ASTM F1614 shall be optimized for user comfort and injury reduction, and shall not exceed 18.0 G in the forefoot and shall not exceed 14.0 G in the rearfoot. Shock attenuation shall be achieved through the choice of standard materials and construction techniques used in commercial industry and found in this Purchase Description. Human biomechanical gait parameters while wearing combat boots shall not be compromised to achieve increased shock attenuation. Testing shall be performed on the completed boot with the removable cushion insert when tested as specified in 4.4.4.

3.3.5 Bond strength. The bond achieved between the boot upper, cushion midsole, and outsole shall be resistant to any delamination for the life of the item.

3.3.5.1 Entire mid-sole to upper. The bond strength shall not be less than 130 pounds. Testing shall be performed on the end item as specified in 4.4.4.

3.3.5.2 Outersole to cushion midsole bond strength. The outersole to cushion midsole bond strength shall not be less than 100 pounds. Testing shall be performed on the end item as specified in 4.4.4.

(NOTE: The components of the sole shall be layered in such a manner that the specified Bond Strengths will be achieved. If test results find values to be less than specified due to material failure in regard to the bond between the layers, the test will be classified as a failure.)

3.3.6 Insole board foam. Microcellular polyurethane foam, 0.125 ($\pm$ 0.01) inch thick, shall be permanently attached to the top of the insole board. The foam shall meet the requirements of Table VIII when tested as specified in 4.4.2.

TABLE VIII. Insole board foam requirements

Characteristics	Requirement
Density, lb/cu.ft	19 - 23
Tensile Strength, psi (min)	60
Compression set, Percent (max) At 50% compression, 70 °C	5
Resilience/Ball Rebound (min)	8

3.4 Boot design. The boot shall be Desert Sand in color and have an all leather upper, breathable, waterproof lining, and direct attached solid rubber sole and polyether polyurethane cushion midsole. Each pair of boots will include two (2) pairs of removable insulated booties. (See Figures 2a and 3 for upper pattern and soling system design requirements.)

3.4.1 Height. The height of the finished boot, measured upward on the outside from tread area at breast of heel, from the bottom of the outsole to the top of boot shall be 11 inches ($\pm$ 0.25) inches on size 10R and shall graduate up and down between sizes and widths.

3.4.2 Boot weight. The weight of a finished boot (size 10R) with cushion insert and bootie shall be less than or equal to 2-1/2 pounds per boot when tested as specified in 4.4.4.

3.5 Lasts and molds The boots shall be made on commercial last model “FMT U3813-1 Military” or equal based on Government approval. An electronic table of last dimensions as well as last bottom files for each size and 3-dimensional data file for sizes 10N (B), 10R (D), 10W (EE) and 10XW (EEEE) for the Direct Attach lasts are available from U.S. Army Natick Soldier Research Development and Engineering Center, RDNS-WPC, Kansas Street, Natick, MA 01760-5011. The numerical size of the boot shall correspond with the numerical size of the last used to produce the boot (i.e. a size 10R boot shall be produced on a size 10R last). The Direct Attach last has a common last bottom shared by N and R lasts and a common last bottom shared by W and XW lasts for each whole and half size. The Direct Attach (either Injection Molding or Open Pouring) process shall utilize a separate cushion midsole mold to produce N and R widths for each whole and half size. The Direct Attach process shall use a separate cushion midsole mold to produce W and XW widths for each whole and half size. The total number of molds required to produce the full size tariff shall be fifty eight (58) molds.

NOTE: To compensate for the bootie it may be necessary to upsize the lasts as follows:

Narrow:	+1/2 size on R width
Regular:	+ 1/2 size on W width
Wide:	+1/2 size on XW width
Extra Wide:	+1 size on XW width

3.6 Construction

3.6.1 Cutting uppers. The leather shall conform to the thickness specified in Table IX when tested as specified in 4.4.2. None of the upper parts shall be split or shaved, except for the eyelet stay.

TABLE IX. Thickness of leather uppers

Part	Thickness (ounces)	
	Minimum	Maximum
Vamp	5	6
Quarter	4-1/2	6
Eyelet stay	2-1/2	3-1/2
Gusset/Tongue	2-1/2	3-1/2
Collar	2	3-1/2
Backstay	4-1/2	6

3.6.2 Skiving. Skiving of upper leather parts is permitted.

3.7 Leakage. The finished boots shall be tested for leakage as specified in 4.4.4.

3.8 Marking, permanent identification. With the exception of product demonstration models (PDMs), the Contractor shall permanently mark the correct American and Mondopoint sizes and widths, and the contract number (Example: 04-D-1234) on the inside quarter panel side area. The marking may be in the form of a label and shall be done in accordance with acceptable commercial practices, and it shall not reduce footwear functionality. The boot shall contain a manufacturing tracking lot number. As an option, each boot may contain a sew-in label identifying the waterproof membrane utilized. Component and boot feature hang-tags may be attached to the boots. Embossing of a brand name shall not be permitted. Both the unit packs and the bootie pack will include a Universal Product Code (UPC) symbol for each National Stock Number (NSN) as specified in the contract solicitation. Each symbol must be Human and scanner readable.

Each removable insulated bootie shall be marked with the sizes and widths of the boot it fits, marked appropriately L and R, and marked to indicate “wash in cold water and line dry” and “not for use as a slipper”. The bootie shall fit the whole boot size indicated as well as 1/2 size larger, (e.g. Size 8 bootie will fit a size 8 and 8-1/2 boot).

3.8.1 Instruction/ and care and use label tag. The instruction/label tag is to be tied to the finished boots and the print size shall be 8 to 10 points, 1/8 inch ($\pm 1/32$ inch). The instruction/label tag and care and use information shall be printed using the data specified on Figures 1 and 2. Each item shall be individually bar-coded. The bar-coding element shall be a 13 digit National Stock Number (NSN). The bar-code type shall be a medium to high density, shall be located so that it is completely visible on the item, and that it causes no damage to the item.

3.9 Thermal insulation. While in its completed configuration, including the removable insulated bootie, the system shall have a minimum overall insulation (clo) value of similar (T) or superior (O) to that of the current Tan Intermediate Cold Wet Boot with Removable Liner. Testing shall be conducted as specified in 4.4.4. The definition of similar and superior are as listed in 4.4.4.

3.10 Thermal protection. The boots shall meet the flame, heat and liquid penetration resistance requirements. Testing shall be as specified in 4.4.4.

3.10.1 Flame resistance. The boots shall meet the Flame Resistance Test four requirements of NFPA 1971, 2007 edition when tested as specified in 4.4.4.

3.10.2 Upper heat resistance. The boots shall meet the Thermal Conductive Heat Resistance Test One requirements of NFPA 1971, 2007 edition when tested as specified in 4.4.4.

3.10.3 Sole heat resistance. The boots shall meet the Thermal Conductive Heat Resistance Test Two requirements of the NFPA 1971, 2007 edition when tested as specified in 4.4.4.

3.10.4 Liquid penetration resistance. The boots shall meet the Liquid Penetration Resistance Test requirements of the NFPA 1971, 2007 edition when tested as specified in 4.4.4.

3.10.5 Whole boot Moisture Vapor Transmission Rate (MVTR). The whole boot breathability (MVTR) shall not be less than 3.0 grams/hour when tested in accordance with 4.4.4.

3.11 Workmanship. The finished boots shall conform to the quality of product established by this Purchase Description. The occurrence of defects shall not exceed the contractor's own quality assurance standards and the quality assurance standards defined by the technical data in the bid package. Internal seams shall not cause irritation, "hot-spots", blisters, or other damage to the user.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

- a. First article inspection (see paragraph 4.2).
- b. Conformance inspection (see paragraph 4.3).

4.1.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.4.2, 4.4.4 and 4.5.

4.2 First article inspection. A first article, submitted in accordance with 3.1.1, shall be inspected, examined for appearance, color and finished defects in Table XII and tested for the characteristics as specified in Table XIII.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4.3 and the tests of 4.4.2 through 4.4.4 as applicable. Sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4 and with quality acceptance limits as specified in the contract and/or order, except where otherwise indicated.(see 6.2)

4.4 Inspection and component testing.

4.4.1 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether the components are as specified or operations and/or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2 Component testing. The components shall be tested for the characteristics listed in Table X. The methods of testing specified where applicable shall be used. The sampling plan shall be in accordance with the following unless otherwise specified in the contract or solicitation:

Lot size (pairs)	Sample size
Under 800	2
801 – 22,000	5
22,001 and above	8

TABLE X. Component tests

Material	Characteristics	Requirement paragraph	Test Method
All Exterior Components			
	Shade Evaluation	3.1.3	AATCC Procedure 9 <u>1</u> /
Upper Leather			
	Range of characteristics	3.2.1.1	MIL-PRF-3122
Insoles			
	Thickness	3.2.2	ASTM D1777 <u>2</u> /
	Fungal resistance	3.2.2	ASTM D3273
	Wet Tensile	3.2.2	SATRA TM2
	Flexing Index	3.2.2	SATRA TM3

TABLE X. Component tests - Continued

Material	Characteristics	Requirement paragraph	Test Method
Insoles Cont.	Abrasion Resistance		
	Cellulose board	3.2.2	SATRA TM14
	Non-Woven board	3.2.2	SATRA TM31
	Transverse Tensile-Dry	3.2.2	SATRA TM80
	Dimensional Stability	3.2.2	SATRA TM98
	Peel Strength	3.2.2	SATRA TM101
Non removable boot lining -Laminated Cloth			
	Martindale Abrasion		
	Fabric Only, Dry	3.2.3.1	ASTM D4966 <u>3/</u>
	Bursting Strength		
	Fabric Only	3.2.3.1	ASTM D3787
	Bacterial Resistance		
	<i>Staphylococcus Aureus</i>	3.2.3.1	AATCC 100 & ATCC-6538
	<i>Pseudomonas Aeruginosa</i>	3.2.3.1	AATCC 100 & ATCC-9027
	Hydrostatic Resistance (psi)		
	Liner cloth (initial)	3.2.3.1	ASTM D751 <u>4/</u>
	At Seam	3.2.3.1	ASTM D751 <u>5/</u>
	After Synthetic Perspiration	3.2.3.1	ASTM D751 <u>6/</u> , <u>7/</u> , <u>8/</u>
	After Diesel Fuel Oil	3.2.3.1	ASTM D751 <u>8/</u> , <u>9/</u>
	Water Permeability/ Resistance		
	Liner cloth (initial)	3.2.3.1	AATCC 127 <u>10/</u>
	At Seam	3.2.3.1	AATCC 127 <u>11/</u> , <u>10/</u>
	After Synthetic Perspiration	3.2.3.1	AATCC 127 <u>7/</u> , <u>8/</u> , <u>10/</u> , <u>12/</u>
	After Diesel Fuel Oil	3.2.3.1	AATCC 127 <u>13/</u> , <u>8/</u> , <u>10/</u>
	Abrasive Durability (cycles to leak)		
	Back / Leather Side	3.2.3.1	ASTM D3886 and AATCC 127 <u>14/</u> , <u>10/</u>
	Moisture Vapor Transmission Rate (g/m ² /24hrs)		
	B method	3.2.3.1	ASTM E96 <u>15/</u>
	BW method	3.2.3.1	ASTM E96 <u>16/</u>
	Flex Durability		
	Gelbo (cycles to leak)	3.2.3.1	ASTM F392 and AATCC 127 <u>17/</u> , <u>10/</u>
	Wet (hrs to leak)	3.2.3.1	ASTM F392 and AATCC 127 <u>18/</u> , <u>10/</u>

TABLE X. Component tests - Continued

Material	Characteristics	Requirement paragraph	Test Method
Removable Insulated Bootie			
Batting	Composition (polyester %)	3.2.4	AATCC 20A
	Weight (grams/sq.m)	3.2.4	ASTM D3776
	Thermal Resistance R-Value (h.F.ft ² /Btu)	3.2.4	<u>19</u> /
	Water Gain (wt%)	3.2.4	<u>20</u> /
Quilt	Weight grams/sq.m)	3.2.4	ASTM D3776
	Thermal Resistance (clo)	3.2.4	ASTM C518
	Bootie Thickness (in.)	3.2.4	ASTM D1777 <u>21</u> /
Box Toe/Counters			
	Initial Collapsing Load (N)	3.2.5/3.2.6	SATRA TM83
	Resilience (%)	3.2.5/3.2.6	SATRA TM83 <u>22</u> /
	Moisture Resistance (%)	3.2.5/3.2.6	SATRA TM83 <u>23</u> /
	Area Shape Retention (%)	3.2.5/3.2.6	SATRA TM83
Shank		3.2.7	Visual
Speed lace		3.2.8	Visual
Eyelets		3.2.9	Visual
Removable Cushion Insert			
	Density (lbs./cu.ft.)	3.2.10	ASTM D297 or ASTM D3574
	Foam Hardness (Shore 00)	3.2.10	ASTM D2240
	Tensile strength (psi)	3.2.10	ASTM D3574
	Compression set @ 77°F	3.2.10	ASTM D3574
Laces		3.2.11	A-A-55093
Fastener Tape		3.2.12	A-A-55126
Collar Foam			
	Compression (psi)	3.2.13	ASTM D1056
Cushion Midsole (Polyurethane)			
	Density (gm/cc)	3.3.2	ASTM D3574
	Hardness (shore A)	3.3.2	ASTM D2240 <u>24</u> /
	Tensile strength (kgf/cm sq)	3.3.2	ASTM D3574 <u>25</u> /
	Elongation (%)	3.3.2	ASTM D3574 <u>25</u> /
	Tear strength Die T (kg/cm)	3.3.2	ASTM D3574 <u>25</u> /
	Volume swell (%) w/ ASTM Oil #3 (at 24 hrs)	3.3.2	ASTM D471 <u>25</u> /, <u>26</u> /
	Compression Set %	3.3.2	SATRA TM64 <u>27</u> /
	Ross flex (1x6 inch) at 5°C	3.3.2	ASTM D1052 <u>28</u> /, <u>29</u> /

TABLE X. Component tests - Continued

Material	Characteristics	Requirement paragraph	Test Method
Outsole (rubber)			
	NBS abrasion	3.3.3.1	ASTM D1630
	Hardness (shore A)	3.3.3.1	ASTM D2240
	Stitch Tear (lbs.)		
	(dry)	3.3.3.1	ASTM D4786
	(Fuel B)	3.3.3.1	ASTM D4786 <u>30/</u> , <u>31/</u>
	Die C tear strength (lbs.)	3.3.3.1	ASTM D624
	Ross flex, Oil #3 (%)	3.3.3.1	ASTM D1052 <u>32/</u>
	Volume swell (%)		
	Fuel B (@ 46 hrs)	3.3.3.1	ASTM D471 <u>33/</u>
	Non-marking	3.3.3.1	SATRA TM223
	Tensile (lbs.)	3.3.3.1	ASTM D412
	Elongation (%)	3.3.3.1	ASTM D412
	Cold Flex (@ -23°C)	3.3.3.1	ASTM D1052 <u>34/</u>
	Transitional Hardness (2 hrs.@ -23°C)	3.3.3.1	ASTM D2240
	Slip Index (rough /ice)	3.3.3.1	SATRA TM144 <u>35/</u>
Insole board foam			
	Thickness	3.3.6	Visual
	Density	3.3.6	ASTM D3574
	Tensile Strength	3.3.6	ASTM D3574
	Compression set		
	At 50%, 70°C	3.3.6	ASTM D3574
	Resilience, Ball Rebound	3.3.6	ASTM D3574

1/ The color and appearance of all exterior components shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500 (± 200) °K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200) °K.

2/ A thickness gauge of the dead-weight type equipped with a dial graduated to read directly to 0.001-inches shall be used. The presser foot shall be circular, with a diameter of 1.129 (± 0.001) inches and with moving parts weighted to apply a total load of 0.60 (± 0.03) pounds per square inch (psi) to the specimen. The anvil shall be not less than 1.129-inches in diameter. The presser foot and anvil surface shall be planed to within 0.001 inches and shall be parallel to each other to within 0.001-inches.

3/ As described in the test method; Martindale Tester; Pressure applied: 12 kPa; visually examine specimen after each 3200 cycles; abradant changed after each 3200 cycles. Use Option 1 for evaluation; no hole appears.

4/ Test five (5) samples in accordance with ASTM D751 Procedure A1 with the face (foot side) of the liner cloth in contact with the water.

5/ Test five (5) samples containing sealed seams in accordance with ASTM D751 Procedure A1 with the face (foot side) of the liner cloth in contact with the water and the seam centrally located in the circular orifice.

6/ Test five (5) samples, eight inch by eight inch (8" x 8"), cut from the laminate material and exposed to contamination, synthetic perspiration (see footnote 7/). Contaminate prior to leak testing by dispensing 2 ml of perspiration solution on both sides of the sample (see Footnote 8/ for contamination procedure). Determine hydrostatic resistance in accordance with ASTM D751 (Procedure A1) with the face (foot side) of the liner cloth in contact with the water.

7/ Synthetic perspiration shall be made up in a 500 ml glass beaker by combining 3.0 grams sodium chloride, 1.0 gram trypticase soy broth powder, 1.0 gram normal propyl propionate, and 0.5 grams of liquid lecithin. Add 500 ml of distilled water, add a magnetic stirring bar and cover beaker. Place the beaker on a combination hot plate/ magnetic stirring apparatus. While stirring, heat the solution to 50°C until all ingredients are dissolved. Then, while stirring, cool solution to 35°C.

8/ Dispense contaminate on to the center of a glass plate (8 inches by 8 inches by 0.25inch thick.). Sample shall be placed face down on the plate and an additional contaminate dispensed on the backside. Place a second glass plate on the sample. Next, position a 4-pound weight on the center of the second glass plate. After 16 hours, remove the sample (do not rinse) and test immediately.

9/ Test five (5) samples, eight inches by eight inches (8" x 8"), cut from the laminate material and exposed to diesel fuel as follows: Contaminate prior to leak testing by dispensing 3 drops of diesel fuel onto both sides of the sample (see Footnote 8/ for contamination procedure.) Determine hydrostatic resistance in accordance with ASTM D751 Procedure A1 with the face (foot side) of the liner cloth in contact with the water.

10/ Test five (5) samples in accordance with AATCC 127 with face/ foot side towards water. Suter test conditions shall be 0.7 psi (50 cm), for 5 minutes. Any observed water droplets in test area constitutes a failure.

11/ Test five (5) samples containing sealed seams in accordance with AATCC 127 with the face (foot side) of the liner cloth in contact with the water and the seam centrally located in the circular orifice. Follow Suter test conditions and inspection as specified in Footnote 10/.

12/ Test five (5) samples, eight inches by eight inches (8" x 8"), cut from the laminate material and exposed to synthetic perspiration (Footnote 7/.) Contaminate prior to leak testing by dispensing 2 ml of perspiration solution on both sides of the sample (see Footnote 8/ for contamination procedure.) Determine water resistance in accordance with AATCC 127 with the face (foot side) of the liner cloth in contact with the water. Follow Suter test conditions and inspection specified in Footnote 10/.

13/ Oil Test five (5) samples, eight inches by eight inches (8" x 8"), cut from the laminate material and exposed to diesel fuel as follows: Contaminate prior to leak testing by dispensing 3 drops of diesel fuel onto both sides of the sample (see Footnote 8/ for contamination procedure.) Determine water resistance in accordance with AATCC 127 with the face (foot side) of the liner cloth in contact with the water. Follow Suter test conditions and inspection as specified in Footnote 10/.

14/ Test back side of five (5) laminate samples in accordance with ASTM D3886. Non-contact pin diaphragm approved for use to decrease variability. The test is to be run for the specified number of cycles prior to testing for leakage. Leakage shall be determined in accordance with AATCC 127. Follow Suter test conditions and inspection as specified in Footnote 10/.

15/ Determine MVTR in accordance with ASTM E96 method B with the back side of the test cloth shall face the water. The free stream air velocity shall be 550 ($\pm$ 50) fpm as measured 2-inches above the specimen. The airflow shall be measured at least 2-inches from any other surface. The test shall be run for 24 hours and weight measurements shall be taken at only the start and completion of the test. At the start of the 24-hour test period, the air gap between the water surface and the specimen shall be 0.75 ($\pm$ 0.06) inches.

16/ Determine MVTR in accordance with ASTM E96 method BW with the backside of the test cloth facing the water. The free stream air velocity shall be 550 ($\pm$ 50) fpm as measured 2-inches above the specimen. The airflow shall be measured at least 2-inches from any other surface. The specimen shall be sealed to the cup in any manner that prevents wicking or leaking of water out of the cup. The test shall be run for 2 hours and weight measurements shall be taken at only, the start and completion of the test.

17/ One sample, 8 by 12 inches., shall be conditioned and flexed in accordance with ASTM F392 except specimen shall not be aged, the short edges shall not be heat sealed or otherwise joined, and the specimen shall be flexed for the specified number of cycles. Two (2), 6 by 8 inch specimens shall be cut from the flexed sample and tested for leakage in accordance with AATCC 127. Follow Suter test conditions and inspection as specified in Footnote 10/.

18/ One sample, 14 inches by full width, shall be selected from a sample unit and tested for leakage after the required number of hours of continuous agitation. The specimen shall be agitated using the “normal,” cycle in an automatic home laundering machine except that the washing machine shall be capable of continuous agitation. The water level shall be maintained at 16 ($\pm$ 0.5) gallons, and the water temperature shall be 32 ($\pm$ 9) $^{\circ}$ C. Additional fabric shall be added to the specimen to create a load weight of 2 ($\pm$ 0.2) pounds. Upon completion of the continuous agitation the specimen is to be air dried and then tested for leakage in Accordance with AATCC 127. Follow Suter test conditions and inspection as specified in Footnote 10/.

19/ The thermal resistance should be tested using a hot plate thermal conductivity analyzer under a pressure of 0.290psi. The plate temperatures should be set at 50 $^{\circ}$ F (lower) and 100 $^{\circ}$ F (upper) resulting in a temperature differential of 50 $^{\circ}$ F.

20/ The sample (15cm x 15cm) is conditioned for 24 hours in a climate of 20 $^{\circ}$ C (68 $^{\circ}$ F) and 65% RH and weighed. Then the sample is submerged for 5 minutes in distilled water and afterwards rotated in a centrifuge for 23 seconds at 1500rpm. After this, the sample is again submerged for 5 minutes in distilled water and again rotated 23 seconds. Then the sample is taken out of the centrifuge and its water uptake (delta G) is determined by weighing.

21/ The thickness of the insulated bootie shall be measured in accordance with ASTM D1777 with a sample size of 2 x 2 inches.

22/ Resilience is percentage retention of initial collapsing load after ten (10) collapses.

23/ Moisture resistance is percentage retention of initial dry collapsing load after 1 hour immersion.

24/ Thickness of test slab shall be 1/2-inch as stated in the respective ASTM method.

25/ Thickness of test sample used shall equal 12.7mm (0.5 in) from test slabs or 6.0mm from actual cushioned midsoles. Actual cushioned midsoles shall be used for verification testing only. However, all test reports used for conformance purposes and submitted to the Government shall use test slabs.

26/ Test conducted after 24 hours at Room Temperature with ASTM test slabs.

27/ Thickness of test sample used shall equal 12.7mm (0.5 in) from test slabs or 19.0mm from actual cushioned midsoles. Actual cushioned midsoles shall be used for verification testing only. However, all test reports used for conformance purposes and submitted to the Government shall use test slabs.

28/ Method modified to include Texon board (2mm) glued onto the specimens and conducted at -5°C. Samples shall be preconditioned for 2 hours at -5°C.

29/ Thickness of test sample used shall equal 6.3mm (0.25 in) from test slabs or 6.0mm from actual cushioned midsoles. Actual cushioned midsoles shall be used for verification testing only. However, all test reports used for conformance purposes and submitted to the Government shall use test slabs.

30/ Test conducted after 46 hours in Fuel B at Room Temperature.

31/ Conditioned for 2 hours, heat aging not necessary

32/ One (1) drop of oil per 6,000 flexes.

33/ Test conducted after 46 hours at Room Temperature with ASTM test slabs

34/ Test conducted after 2 hours conditioning

35/ Test in accordance with frosted ice method, flat mode. Sole to be pre-conditioned with 40 grit sand paper for six (6) cycles. Test conducted on end item.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in Table XII below. The lot size shall be expressed in units of boots. The inspector shall check to see that cushion inserts are inserted in the boots. For the pairing examination, and when determining possible differences in outersole thickness, the pair shall be examined together. During the inspection for the defects listed within the "End item visual defect table", the departures from specification requirements listed in the table shall be observed and recorded. Presence of these departures shall not result in the rejection of the lot, but the plant quality assurance and Government personnel reserve the right to establish and impose lot rejection criteria when either the severity or the presence of these departures warrant. These additional rejection criteria may be invoked at any time during the life of the contract when the Government judges this action to be in their best interest. Defects of pairing shall be classified as a single defect.

NOTE: Appearance defects shall be scored only when the condition is plainly visible at a distance of 3-feet or more and it appreciably affects item appearance. Conditions classified as (*) conditions shall be scored as major defects when seriously affecting serviceability or appearance, and as a minor departure when slightly affecting serviceability or appearance.

TABLE XII. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Pairing	Not properly mated; i.e., not right and left of same size	101	
	Variation in color, luster, or appearance	*	
	Variation of more than 1/4-inch in height of pair	102	
	Box toe malformed	*	
	Difference in outersole thickness between left and right boots - more than 3/32-inch - more than 1/16-inch but not more than 3/32-inch	103	201
Cleanness	Any non-removable spot, stain, or foreign matter affecting appearance		202
Color and Finish	Color not as specified	*	
Design	Not as specified	104	
Construction and Workmanship (General)	Any cut, tear, hole, repair or damage	*	
	Any component or assembly omitted or misplaced, operation omitted, not properly performed.	*	
	Any component, assembly, or design present creating "hot spots"	105	
Seams and Stitching (upper)	Open seam not repaired	106	
	Repair of open seam not as specified		203
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur. On multiple stitched seams, a seam is considered open when either one or both sides of the seam are open. When above conditions occur on the inside of the boot it shall be scored as a * defect.		
	Tight tension resulting in puckering or cutting of the leather	*	
	Loose tension resulting in a loosely secured seam	*	
	Stitching omitted where required	*	
	Gauge of stitching irregular	*	
	Thread ends not trimmed throughout the boot		204
	Needle holes or needle chew	*	
Outersole and Heel	Thickness not as specified	107	
	Any malformation, wrong size used	*	
Eyelets/ Speedlaces	Number of eyelets/speedlaces not as specified in each row	108	
	Not the same number of eyelets/ speedlaces in each row	109	
	Eyelets/speedlaces not properly spaced within the row or misalignment between the rows to an extent interfering with proper lacing	*	
	Any eyelet/speedlace not securely clinched		205
	Any eyelet/speedlace malformed or damaged		206

TABLE XII. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Marking and instruction tag	Missing, incomplete, incorrect, not applied in the specific manner, misplaced, illegible, or not specified size		207
Removable Liner	Torn or missing	110 *	
	Not properly mated with boot		208
Cushion insole (insert)	Missing, wrong size, or wrong side up	111	
Lining	Torn, loose or wrinkled	112*	
Upper Leather	Not flesh side out	113	
	Leather deeply scuffed/damaged		*
	Grub or tick damage, scratches, brands or boney leather		*
	Fat wrinkles or veins	114	
	Thickness exceeds maximum or less than minimum specified	115	
Gusset, tongue and collar leather	Flanky		*
	Loose flesh, boney or boardy		*

4.4.4 End item testing. The finished boot shall be tested for the characteristics listed in Table XIII. All test results shall contain the individual values utilized in expressing the final results. For all item tests the sample size shall be five (5) boots. The number of determinations shall be a specified in the contract or solicitation. The lot will not be acceptable if one or more samples fail any test.

TABLE XIII. End item testing

Material	Characteristics	Requirement paragraph	Test Method
Whole boot			
	Shock attenuation	3.3.4	ASTM F1614
	Bond Strength	3.3.5	
	Entire midsole to upper	3.3.5.1	4.5.2
	Outersole to cushion midsole	3.3.5.2	4.5.1
Boot Design and Construction			
	Height	3.4.1	Visual
	Weight	3.4.2	4.4.4
	Thickness	3.6.1	Visual
	Leakage	3.7	4.5.3
	Thermal insulation	3.9	4.5.5 <u>1/</u>

TABLE XIII. End item testing - Continued

Material	Characteristics	Requirement paragraph	Test Method
Boot Design and Construction - continued			
	Thermal protection	3.10	4.5.6
	Flame resistance	3.10.1	4.5.6.1
	Upper heat resistance	3.10.2	4.5.6.2
	Sole heat resistance	3.10.3	4.5.6.3
	Liquid penetration resistance	3.10.4	4.5.6.4
	Whole boot MVTR	3.10.5	4.5.4

1/ PDM testing only. The Government reserves right to draw samples to test for conformance to this specification at any time.

4.5 Methods of inspection

4.5.1 Bond strength test of outersole to cushion midsole.

4.5.1.1 Specimen. The specimen shall be a completed boot which has aged at least two (2) days. The outersole shall be separated from the cushion-midsole, for a distance of approximately 2-1/2-inches from the toe end of the specimen. The separation may be aided by any suitable device or machine capable of withstanding the amount of force necessary to accomplish separation. After separation, the toe cap of the boot shall be crushed to facilitate mounting of the specimen in the tester. In event that the point of separation at the 2-1/2-inch point is not between the outersole and cushion midsole, that portion of the outersole remaining adhered to the cushion midsole shall be separated manually (with the aid of a knife or other instrument), so as to show separation between the outersole and cushion midsole.

4.5.1.2 Apparatus. A power-driven adhesion machine, or an approved testing device of equal performance, shall be used. The rate of travel of the power-actuated grip shall be 2-inches per minute. The machine shall be operated with a device for maintaining maximum load indication.

4.5.1.3 Procedure. The separate toe ends of the specimen shall be clamped in the jaws of the machine. The specimen shall extend outward at right angles to the direction of the application load. The machine shall be started, and the surface being tested shall be pulled apart to a distance of not more than 4-1/2-inches from the toe. The maximum value attained during separation shall be taken as the bond strength.

4.5.2 Bond strength test of entire soling system to upper.

4.5.2.1 Specimen. The specimen shall be a completed boot which has aged at least 2 days. The specimen shall be legibly marked on each side at two points (2-inches and 4-inches from the tip of the toe). The entire soling system shall then be separated from the upper to the 2-inch

mark. The separation may be aided by any suitable device or machine capable of withstanding the amount of force necessary to accomplish separation. After separation to the two inch mark is achieved, the toe cap of the boot shall be crushed to facilitate mounting of the specimen in the tester. In event that the point of separation at the 2-inch mark is not between the upper and the soling system, that portion of the soling system remaining adhered to the upper shall be separated manually (with the aid of a knife or other instrument), so as to show separation between leather and soling system. Care should be taken in order to be sure that the leather is not cut.

4.5.2.2 Apparatus and procedure. The apparatus for measuring soling system/upper separation shall be as described in test method ASTM D2208. The separated and crushed toe portion of the boot shall be gripped by one clamp in a manner such that the minimum effective jaw surface area applied shall be 1-inch by 1-inch. The jaw surface shall be centered approximately 1/2-inch to 1-inch from the tip of the toe. The other jaw surface shall hold the overlay portion of the separated upper, and it shall not be less than 1-inch by 2-inches, with the long dimensions perpendicular to the pull of the machine. The machine shall be set in operation and the separation continued at a speed of 10-inches per minute until the outsole is separated from the upper to a point past the mark 4-inches from the tip. The maximum value attained during separation shall be taken as the bond strength.

4.5.3 Boot leakage test. The boot leakage test shall be designed to indicate leakage by SATRA TM77, modified as indicated below) while the boot is submerged. The minimum standard for passing this test is 100,000 flexes. Each boot shall be preconditioned by flexing in accordance with SATRA TM 77, dry with a foot form of steel shot for 100,000 flexes.

4.5.3.1 Apparatus. SATRA STM 184H Whole Shoe Flexing Apparatus or equal will be modified to allow the end item to be submerged to the depth of the mid-point between the third and fourth eyelet in accordance with NFPA Standard 1971. The mechanism for flexing on any other apparatus shall be the same as the SATRA STM 184H. The sides of the apparatus must be raised to alleviate splashing on the mechanism for flexing.

4.5.3.2 Procedure. Testing will be in accordance with SATRA MTM 77 at 140 (± 10) flexes/minute and 35 degrees for each flex cycle. A modified SATRA STM 184H whole boot flex apparatus or equal must be used so that the water depth required can be achieved. The top of each item must be sealed appropriately so that the water splashing from the test does not enter through the top of the boot. Absorbent blotting paper will be “balled up” and placed within the boot upper so that any water ingress through the top or upper can be determined. A change in the color between the top and the bottom sections of the absorbent blotting paper will indicate water ingress. The steel shot foot-form shall be removed for testing in water. The end item shall be checked at 100,000 cycles for leakage, as indicated by a color change in the absorbent blotting paper within the boot.

4.5.3.3 Method of inspection. End items will be evaluated in accordance with SATRA TM77, by checking for a change in color of the absorbent blotting paper that is “balled up” and placed within the boot upper. If water enters the top of the boot during the test, new absorbent blotting paper shall be inserted for the remainder of the test, unless the water has leaked to below the fourth eyelet.

If water has definitely leaked through the top of the boot by splashing and reached to below the fourth eyelet, then the whole boot must be dried and the absorbent blotting paper replaced before the test is resumed. Water ingress into the boot by way of wicking up the boot upper (outside) and down into the boot lining (inside) will not be considered an item failure. If it cannot be determined if the water has entered through the top or upper of the boot, a new boot shall be requested and subjected to the test.

4.5.4 Whole boot breathability. The boot breathability test shall be designed to indicate the Moisture Vapor Transmission Rate (MVTR) through the boot by means of a difference in concentration of moisture vapor between the interior and the exterior environment.

4.5.4.1 Apparatus.

- a. The external test environment control system shall be capable of maintaining 23 (± 1)°C and 50 (± 2) percent relative humidity throughout the test duration.
- b. The weight scale shall be capable of determining weight of boots filled with water to an accuracy of (± 0.01) gram.
- c. The water holding bootie insert (WHBI) shall be flexible so that it can be inserted into the boot and conform to the interior contours; it must be thin enough so that folds do not create air gaps; it must have a MVTR value ranging between 920-990 grams/meter²/24 hour. ASTM E96 method B - Standard Test Methods for Water Vapor Transmission of Materials shall be used to determine acceptability (1); and it must be waterproof so that only moisture vapor contacts the interior of the footwear product rather than liquid water. After every five (5) uses of the WHBI it will be disposed of and replaced.
- d. The water circulating bath system for the boot shall be capable of controlling the temperature of the water uniformly in the boot to 35 (± 1)°C as measured in the toe area of the boot.
- e. The footform assembly used with the water circulating bath system shall have a boot plug oriented 12.5 centimeters (5-inches) as measured from the bottom of the plug surface to the inside sole in the heel area.
- f. The boot plug shall be oval shaped measuring 3-inches by 2.5-inches.
- g. The top of the boot shall be sealed to create an impervious barrier to both liquid water and water vapor.
- h. Boots should be laced to the top of the lacing system. Allow for 1 (± 0.5) inch separation between lateral and medial eyelet stays, with possible exception of the eyelets at the top of boot and boot plug area. Maximize seal security at top of boot.
- i. A stationary 6-inch diameter fan shall be used to create the air current past the boot.
- j. The stationary fan shall be positioned perpendicular to the test surface and raised so the center of the fan is 5-inches from the test surface.
- k. The air current origin shall be 15.5-inches from the back heel edge of the boot (D). Refer to Diagram 1.

- l. The air current shall be 250 (± 30) feet/minute at the heel edge of the boot (S). Measurement shall be taken without the boot assembly in place 2-inches up from the test surface at the fan center.

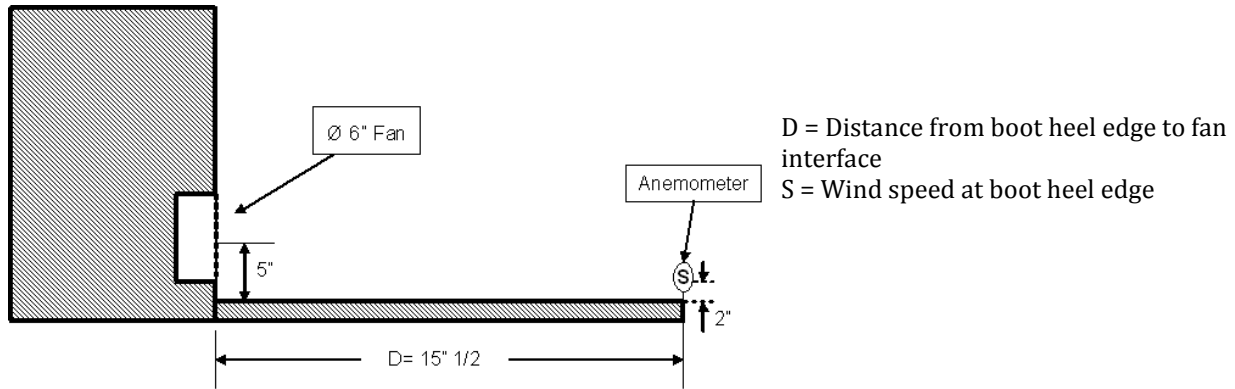


DIAGRAM 1. Whole Boot Breathability Bench-top Set-Up

1/ Determine MVTR in accordance with ASTM E96 method B with the knit side of the membrane facing the water. The free stream air velocity shall be 550 (± 50) feet per minute (fpm) as measured 2-inches above the specimen. The airflow shall be measured at least 2-inches from any other surface. The test shall be run for 24 hours and weight measurements shall be taken at only the start and completion of the test. At the start of the 24-hour test period, the air gap between the water surface and the specimen shall be 0.75 (± 0.06) inches.

4.5.4.2 Procedure.

- a. Remove the removable cushion insert from the boot sample.
- b. Weigh boot sample and record (this will be the unconditioned weight).
- c. The boot shall be conditioned in the test environment for a minimum of 12 hours before testing.
- d. Insert WHBI and footform assembly into boot opening and fill with water preheated to 35 (± 1) $^{\circ}$ C to a height of 12.5 centimeters (5-inches) as measured from the inside sole in the heel area and seal opening with boot plug. The water should be in contact with the bottom of the boot plug.
- e. Regulate water temperature in boot at 35 (± 1) $^{\circ}$ C.
- f. Disconnect water circulating system and weigh boot assembly and record as W_i . The water volume in system shall be noted at the time of weighing.
- g. After weighing reconnect water circulating system and maintain the temperature in boot at 35 (± 1) $^{\circ}$ C for 6 hours (± 5 minutes).
- h. The boot assembly shall be oriented such that the boot sole lies flat on the testing surface with the heel furthest from the fan interface and in line with the center of the stationary fan (See Diagram 2).

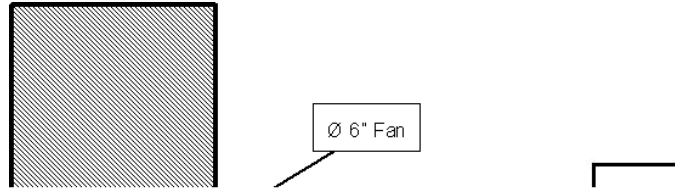


DIAGRAM 2. Whole Boot Breathability Bench-top Set-Up with Boot

- i. After 6 hours (± 5 minutes), reweigh boot assembly. The water volume in system shall match the noted W_i volume. Record weight as W_f and test duration as 6 hours.
- j. Compute whole boot MVTR in grams/hour from the equation below:

$$MVTR = \frac{W_i - W_f}{6}$$

- k. If test is aborted for any reason, thoroughly dry boot to within 5 grams of original boot weight (Step b). Recondition boot sample in test environment for the required length as outlined, and begin test procedure again.

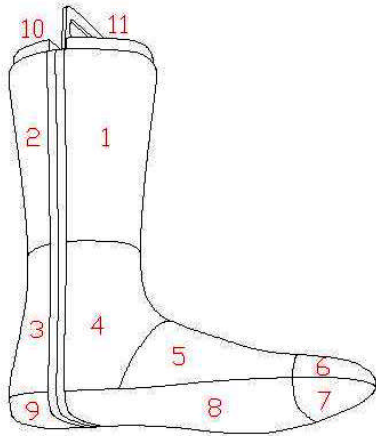
4.5.4.3 Method of inspection. Each boot shall be tested in accordance with the method described in paragraph 4.5.4.2. The average whole boot MVTR from the 6 boots tested shall be greater than 3.0 grams/hour to satisfy the breathability standard.

4.5.5 Thermal insulation.

Test Methodology:

Since there is no ASTM test method to test boots, total thermal insulation of boots is determined using an in-house test methodology, which follows procedures similar to the ASTM F1291 for thermal manikin. The boots encase the 9-zone thermal sweating foot. Tests are conducted in a controlled environmental chamber. Data is collected for thirty minutes of steady state conditions.

The thermal foot provides measurements for segments as shown below.



Thermal Insulation:

Thermal resistance R_t ($m^2 \cdot ^\circ K / Watt$) is a quantity used to determine the insulation value of items. It is a measure of the total resistance to dry heat loss measured on a thermal foot to a relatively cool environment. Clo is one of the most widely used units in the clothing industry. $Clo = 6.45 \cdot R_t$ or $1Clo = 0.155 m^2 K / Watt$

Total Clo refers to the weighted average of all segments of the thermal foot. The toe region consists of segments 6 and 7. The heel region consists of segment 9. The sole region consists of segment 8.

$$R_t = (T_s - T_a) A / H$$

where:

- R_t = total thermal resistance (insulation) of the boot and surface air layer ($^\circ C \cdot m^2 / W$),
- A = surface area of the thermal foot (m^2),
- T_s = temperature at the thermal foot surface ($^\circ C$),
- T_a = temperature in the air flowing over the boot ($^\circ C$), and
- H = power required to heat thermal foot (W).

Test conditions shall be as follow:

Ambient Temperature	15.0 $^\circ C$ (59.0 $^\circ F$) or lower depending on the insulation of the boot
Foot Temperature	35.0 $^\circ C$ (95.0 $^\circ F$)
Wind Speed	0.9 mph (0.4 m/s)

Ideally, three separate samples of the test footwear system are evaluated and the average value is reported. For source selection purposes, one footwear system shall be tested three times and the average value shall be reported. R_t values shall be converted to the more familiar Clo unit for source selection and procurement purposes. Variation of 10% shall be used as the basis for determining similarity, inferiority, or superiority when comparing test items to the standard. determining similarity, inferiority, or superiority between samples. Items that are within $\pm 10\%$ of the insulation provided by the standard/reference shall be considered similar; items that provide less than 90% of the

insulation provided by the standard/reference shall be considered inferior and items that provide more than 110% of the insulation provided by the standard/reference shall be considered superior.

4.5.6 Thermal protection. The boots shall be tested for flame, heat and liquid penetration resistance as specified in paragraphs 3.8.1 thru 3.9.4.

4.5.6.1 Flame resistance. The boots shall meet the requirement of 3.10.1 when tested as specified. The areas of the boot to be tested are in accordance with Figure 8-5.5.2 of NFPA 1971, 2007 edition with the exception of the rubber outsole. The requirement that the boot shall not melt or drip DOES apply, although the boot shall not exhibit any burn-through (reference: Sections 7.10.2 and 8.5). The boot will be considered to have failed the burn-through test when burn-through is observed through all layers of the leather and lining components.

4.5.6.2 Upper heat resistance. The boots shall meet the requirement of 3.10.2 when tested according to NFPA 1971, 2007 edition, sections 7.11.3 and 8.7.

4.5.6.3 Sole heat resistance. The boots shall meet the requirement of 3.10.3 when tested according to NFPA 1971, 2007 edition, section 8.8.

4.5.6.4 Liquid penetration resistance. The boots shall meet the requirement of 3.10.4 when tested according to NFPA 1971, 2007 edition, sections 7.10.4 and 8.28, specific testing to section 9.28.4.2 #4, surrogate gasoline fuel C, as defined in ASTM D471, standard test method for rubber properties, effect of liquids, using a 50% percent by volume of the toluene and iso-octane.)

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The boots are intended for wear by military personnel of the Department of Defense (DoD).

6.2 Acquisition requirements. Acquisition documents should specify the following:

- a. Title, number and date of this Purchase Description.
- b. Sizes required (see 1.2)
- c. The specific issue of individual documents referenced (see 2.2).
- d. When first article inspection is required, (see 3.1, 4.2, 6.3)
- e. Inspection conditions (see 4.1)
- f. Conformance inspection acceptance quality limits (AQLs) (see 4.3)
- g. Sampling plan (if applicable)(see 4.4.2)
- h. Determinations required for end item testing if applicable (see 4.4.4)
- i. Packaging (see 5.1).

6.3 First article. When a first article inspection is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO STANAG 2333. When amendment, revision, or cancellation of this document is proposed that will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization office, if required.

6.6 Equal item. Prior to use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

INSTRUCTION TAG

BOOT, COMBAT, INTERMEDIATE, COLD/WET

FACTS

The Intermediate Cold/Wet Boot (ICWB) with removable liner has been designed to protect your feet from cold injury and frostbite in areas where moisture and cold are critical factors, and where the mean monthly temperature ranges between 14°F and 68°F. The boots should not be worn where temperatures fall below +10°F. (For temperature below +10°F, the Cold Weather, Insulated, Rubber, Black or White, Vapor Barrier Boots should be worn.)

If your feet begin to get cold, button up all your garments and exercise the entire body by swinging the arms, stamping the feet, and rapid movement of the toes.

TRENCH FOOT AND FROSTBITE are serious cold injuries that can cause painful and permanent disability. These injuries can be prevented by proper care of your feet. No boot will keep your feet from freezing in cold/wet conditions if you remain motionless for several hours. Wear the ICWB properly and DO NOT BECOME A COLD WEATHER CASUALTY.

FIGURE 1. Instruction Tag, Boot, Combat, Intermediate, Cold/Wet facts

Use and Care Instructions
Intermediate Cold Wet Boot with Removable Liner

1. Wear with: 1 pair of socks, wool, cushion sole.
2. Sizes. The boots are supplied in whole and half sizes 3 through 15, and in widths, N, R, W, and XW. In some instances it may be necessary to select a slightly larger size than normally worn in order to allow for normal swelling of the feet.
3. Pull on the removable insulated bootie and then the boot, seating the heel firmly in place, then lace. Boots should fit snugly but not tightly. There should be approximately 1/2-inch additional length at toe.
4. Carry extra socks. Change socks often, at least once daily. Remove boots and dry feet as often as possible. Neglect of feet can make you a casualty.
5. Your Combat Boots are designed to be easy to care for. To clean your Combat Boots, brush with stiff nylon bristle brush to clean and then use warm water. Do not use soap to clean your boots. If additional, more stringent cleaning is necessary, only water-soluble cleaning products should be used as oil- or alcohol-based cleaning products may damage your boots. Waterproofing is not required.
6. The removable insulated bootie should be washed in cold water and line dried. The removable insulated bootie is not intended to be worn as a slipper.

FIGURE 2. Use and Care instruction for Boot, Combat Intermediate Cold/ Wet with removable liner

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FIGURE 3A. Boot Design



FIGURE 3B. Removable insulated bootie configuration

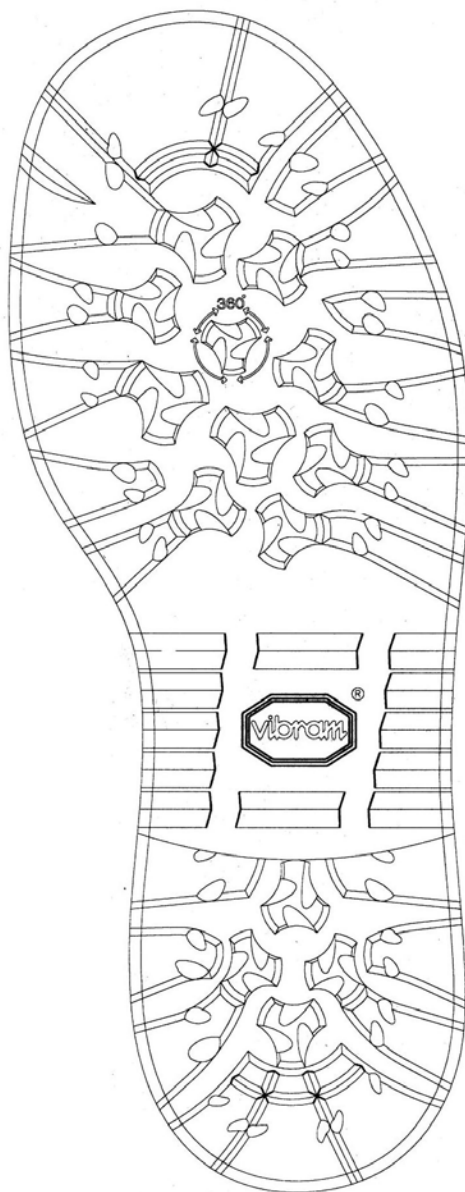


FIGURE 4. Outsole design

Custodians:
Army - GL

Preparing Activity:
Army-GL